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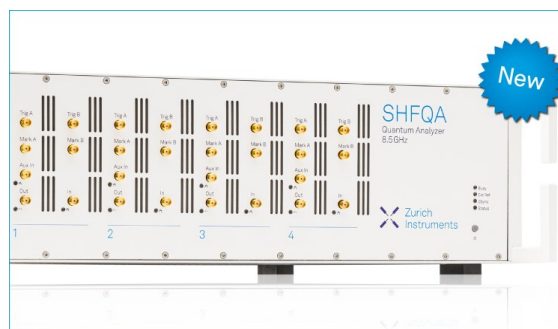
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Oxidation Behavior of $\text{ZrB}_2\text{--ZrC--SiC--ZrO}_2$ Ceramics

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Abstract. The oxidation behavior of ceramic composite materials $\text{ZrB}_2\text{--SiC--ZrC--ZrO}_2$ obtained by sintering under pressure was studied. It was found that the introduction of ZrO_2 in the composition of ceramic composites leads to their compaction. An increase in the content of zirconium dioxide in ceramics based on ZrB_2 intensified the oxidation process of the surface of composites, which can be demonstrated by the increase in sample mass observed during heating, which has a positive effect on the self-healing of the already formed defects.

INTRODUCTION

Materials from refractory compounds based on carbides, borides, and nitrides of transition metals of groups IV and V of the chemical element periodic system with melting points above 3000°C are classified as ultra-high temperature ceramics (UHTC) [1–3]. Most of these materials have excellent combinations of properties—high hardness, high elastic modulus, good wear resistance, satisfactory electrical conductivity, and thermal conductivity, good creep resistance, and chemical stability [1–5]. UHTC, as heat-protective and heat-resistant materials, is used in foundries, power engineering, and aerospace industry. However, the range of use of these materials can be much more extensive, as other types of UHTC are very sensitive to defects (cracks, microcracks) [6]. The appearance of defects in ceramic products during operation leads to the development of catastrophic damage, which negatively affects their reliability and durability.

It is known that self-heal of defects and self-repair of properties is possible in some ceramics. The main mechanisms of self-healing in ceramics are diffusion-controlled sintering, viscous containment of propagating cracks, controlled redistribution of the glassy intercrystalline phase, filling of the crack opening space with oxidation reaction products; phase transformations [7–11]. However, each of the listed self-healing mechanisms is implemented within a narrow temperature range. The creation of structural-phase prerequisites for the synergistic action of several self-healing mechanisms of ceramics in a wide temperature range will improve their reliability under thermally loaded conditions and thus expand the field of practical applications. The aim of this work is to obtain and study self-healing in $\text{ZrB}_2\text{--SiC--ZrC--ZrO}_2$ ceramic composites

MATERIALS AND METHODS

For obtaining ceramic composite materials $\text{ZrB}_2\text{--SiC--ZrC--ZrO}_2$, zirconium diboride, zirconium carbide, silicon carbide, and zirconium dioxide powders were used as the main components. The ratio of ZrB_2 , ZrC , and SiC

components in the ceramic matrix was constant for all powder mixtures, and the volume ratios were 60, 15, and 25% respectively. The content of zirconium dioxide was 0, 5, 10, 15 vol %

Preparation of powder mixtures with simultaneous mechanical activation was carried out in an AGO planetary ball mill in argon atmosphere for 5 minutes at a rotation frequency of 1820 rpm. Ceramic composite samples were obtained by powders hot pressing at a 1900°C and a pressure of 30 MPa for 10 min in an argon atmosphere. The samples had the shape of disks $\varnothing=10\pm0.3$ mm, $h=3\pm0.3$ mm.

The structure of the composites was studied by a scanning electron microscopy Vega Tescan 3 microscope. EDS analysis was performed using a nitrogen-free ADD detector Inca x-ACT. X-ray phase and X-ray diffraction studies were carried out using X-ray diffraction patterns using $\text{CuK}\alpha$ radiation.

Cut-shape defects on the samples surface were applied by blade with a diamond paste. The depth was 170 ± 25 μm . Self-healing of defects in ceramic composites was studied by successive heating of samples to temperatures of 1200, 1400, 1600°C in an air atmosphere with a heating rate of 250 °C/h. Self-healing of defects during thermal exposure is characterized by the percentage of self-healing, calculated from the ratio of the residual defect depth after thermal exposure to the initial defect depth: $H=(L_0-L_n)/L_0\times100\%$, where L_0 is the depth of the initial defect, L_n is the depth defect after thermal exposure [12].

RESULTS AND DISCUSSION

The phase composition of $\text{ZrB}_2\text{-SiC-ZrC}$ composites was represented by hexagonal ZrB_2 and SiC, cubic ZrC. Reflections corresponding to the tetragonal modification of ZrO_2 were also recorded on X-ray diffraction patterns of $\text{ZrB}_2\text{-SiC-ZrC-ZrO}_2$ composites. After heating to 1200°C, the phase composition on the surface of $\text{ZrB}_2\text{-SiC-ZrC}$ composites was only represented by the monoclinic modification of zirconium dioxide. The X-ray diffraction patterns of the $\text{ZrB}_2\text{-SiC-ZrC-ZrO}_2$ composites after heating, in addition to monoclinic ZrO_2 reflections, recorded peaks of the tetragonal modification of zirconium dioxide. An increase in the volume content of ZrO_2 led to a change in the composites' structural characteristics. The size of the coherent diffracting domain (CDD) of m- ZrO_2 was equal to 30 nm in the $\text{ZrB}_2\text{-SiC-ZrC}$ ceramics after heating; the CDD value of m- ZrO_2 was equal to 50 nm in the $\text{ZrB}_2\text{-SiC-ZrC-15 vol \% ZrO}_2$.

Figure 1 shows the structure of $\text{ZrB}_2\text{-SiC-ZrC-15 vol \% ZrO}_2$. According to the measurement, an increase in the zirconium dioxide content led to a decrease in the average grain size of the composites. Thus, the average grain size of the $\text{ZrB}_2\text{-SiC-ZrC}$ composite was 4 ± 0.14 μm . In the composite with a content of 15 vol % ZrO_2 , the average grain size was 2.4 ± 0.1 μm . The EDS analysis of the ceramic composites samples surface after heating to 1200°C showed that the oxide layer contained O, Zr, Si, and B atoms. According to [2, 4], during oxidation, the layer consisting of a porous skeleton is formed on the surface of ZrB_2 -based ceramics, formed by branched white crystals of zirconia dioxide surrounded by borosilicate glass, which is represented as dark areas in SEM images in BSE mode. The surface of the composite materials was oxidized after annealing to 1200°C. X-ray studies showed that as a result of heating, the phase composition on the surface of the composites changed and was represented mainly by a monoclinic modification of zirconium dioxide.

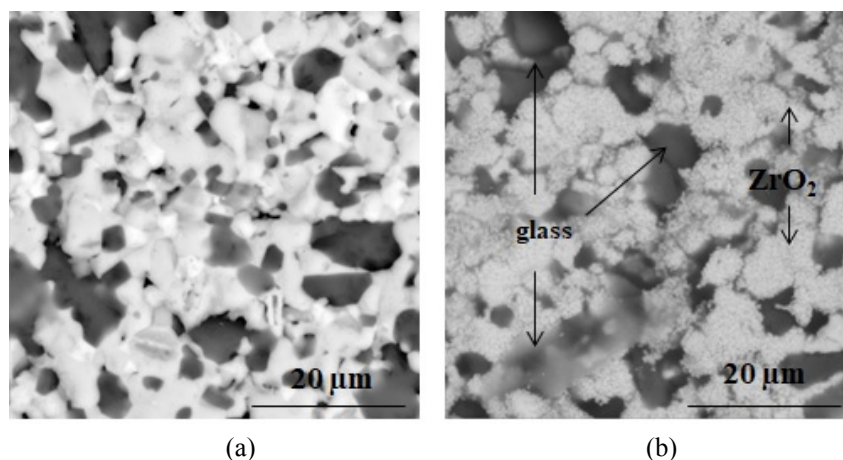


FIGURE 1. SEM-image of the composite material surface $\text{ZrB}_2\text{-ZrC-SiC-15 vol \% ZrO}_2$: (a) initial and (b) after heating to a temperature of 1200°C.

The mechanism of self-healing of defects in ZrB_2 -based ceramics is based on oxidation processes. When interacting with air, zirconium diboride is already oxidized at 800°C to form B_2O_3 and ZrO_2 , which leads to an increase in the mass of ceramic samples [13, 14]. Due to the high vapor pressure, B_2O_3 evaporates at a temperature above 1100°C at atmospheric pressure [15]. In this case, a porous ZrO_2 layer with a columnar microstructure is formed on the surface of the material, which increases oxygen transport to the reaction boundary and provokes significant oxidation of the material [16]. When silicon-containing additives, for example, SiC , are introduced, borosilicate glass forms on the borides surface, which acts as an effective barrier to the rapid transfer of oxygen into the material body. At temperatures above 1400°C , silicon carbide is actively oxidized, and SiO_2 is formed under the glass phase layer. In this case, the formation of the porous depleted SiC layer under the oxide layer is possible, facilitating the diffusion of oxygen through the oxide layer to the non-oxidized matrix [2, 10–18]. It was reported in [19] that the addition of ZrC prevents the formation of the depleted SiC layer.

Backscattering images (BSE) of defects formed on the surface of ZrB_2 - SiC - ZrC samples oxidized at various temperatures are shown in Fig. 2. It can be seen that during the oxidation process, the formed defects were filled with ceramic oxidation products, which indicates the healing of the defects. In this case, for ZrB_2 - SiC - ZrC ceramics, oxidized to a temperature of 1200°C , the defect was filled mainly with borosilicate glass (dark region), Fig. 2b. With the increase of temperature, the contribution of the oxide layer to the healing of defects increased. Defects formed on the surface of the ZrB_2 - SiC - ZrC composite, oxidized at 1600°C , and were completely filled with oxidation products, mainly crystalline oxides.

Oxidation of samples at a temperature of 1200°C is characterized by the maximum mass gain compared with other temperatures. In addition, for all temperature tests, there is a tendency to increase the mass gain of samples as the volume content of zirconium dioxide increases. The change in the mass of samples under high-temperature influences can be caused by the following factors: active oxidation of the components that make up the ceramic composites, and the evaporation of low-melting compounds.

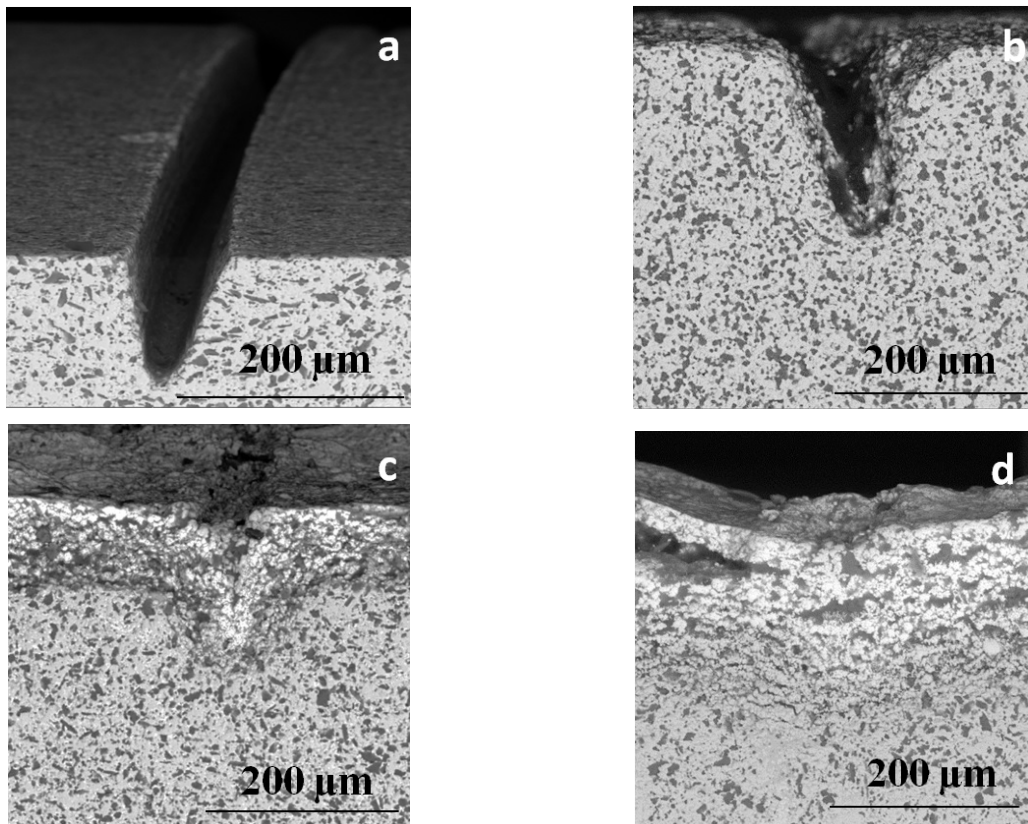


FIGURE 2. SEM image of the cross section of a defect formed on the surface of a ZrB_2 - SiC - ZrC composite oxidized at 25 (a), 1200°C (b), 1400°C (c), 1600°C (d).

During the oxidation process at a temperature of 1200°C, a layer of borosilicate glass forms on the surface of the composites, which acts as a barrier for the oxygen transport to the interior of the material body. An increase in the volume content of ZrO₂ facilitates the process of oxygen diffusion, which contributes to the active oxidation of composites and causes a significant increase in the mass of ZrB₂-SiC-ZrC-ZrO₂ composites. When the temperature rises above 1400°C, the process of glass phase formation intensifies, which increases the oxidation resistance of ceramics. According to the triple-phase diagram of ZrO₂-B₂O₃-SiC, the composite oxidizes when heated to 1600°C to form a complex oxide consisting of the liquid phase (B₂O₃ + SiO₂) and the solid ZrO₂. During the evaporation of boron oxide, the silicate rich glass phase is formed. In this case, an oxide layer is formed on the ceramic surface, formed by the ZrO₂ skeleton and the glass phase layer. In general, more effective self-healing of defects occurred during the oxidation of ceramics to a temperature of 1600°C, where the formed defects were completely filled by the mixture of ceramic components oxidation products, the percentage of self-healing was 100% for all compositions of ZrB₂-SiC-ZrC-ZrO₂ composites. Ceramic ZrB₂-SiC-ZrC-ZrO₂ oxidation at the temperature of 1400°C provided self-healing of defects by an average of 60–100%, depending on the amount of zirconia additive. Therefore, for ceramic ZrB₂-SiC-ZrC, H ~ 60%. The introduction of 15 vol% ZrO₂ additives into the composite resulted in complete healing of defects. Oxidation at 1200°C led only to partial self-healing of defects in composites of all compositions. However, the self-healing percentage of defects in the ZrB₂-SiC-ZrC composite samples did not exceed 50%, and for a composite, with a content of 15% zirconia, it was 80%.

CONCLUSION

Thus, in this work, the oxidation behavior of ZrB₂-SiC-ZrC-ZrO₂ obtained by sintering under pressure was studied. An increase in the content of zirconium dioxide in ZrB₂-based ceramics intensified the oxidation process on the surface of the composites, as evidenced by the increase in the oxide layer thickness on the surface of the samples and the increase in the mass of the samples observed during heating, which positively affected the self-healing of the formed defects. Oxidation of ceramics at 1600°C resulted in the complete healing of defects, regardless of the content of ZrO₂. Oxidation of ceramics ZrB₂-SiC-ZrC-ZrO₂ at 1400°C provided defect self-healing by an average of 60–100%, depending on the amount of zirconia additive. The most effective healing of defects (H = 80%) was observed in ZrB₂-SiC-ZrC-15 vol% ZrO₂ ceramics oxidized at 1200°C.

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REFERENCES

1. A. Paul, et al., *J. Am. Ceram. Soc.* **91**, 22–28 (2012).
2. R. Inoue, et al., *J. Mater. Sci.* **53**(21), 14885–14906 (2018).
3. L. M. Rueschhoff, et al., *Ceram. Eng. Sci. Proc.* **2**(1), 22–37 (2020).
4. N. L. Savchenko, et al., *Wear* **446**, 203204 (2020).
5. A. G. Burlachenko, et al., *Russ. Phys. J.* **62**(8), 1438–1444 (2019).
6. Sudakshina Dutta and J. M. Chandra Kishen, *Phys. Mesomech.* **22**(2), 96–106 (2019). doi 10.1134/S1029959919020024
7. S. van der Zwaag, *Self Healing Materials: An Alternative Approach to 20 Centuries of Material Science* (Springer, 2008), pp. 1–18.
8. B. Aïssa, et al., *Adv. Mater. Sci. Eng.* **2012**, 1–17 (2012).
9. M. Mobaraki, et al., *Self-Healing Composite Materials* (Woodhead Publishing, 2020), pp. 15–31.
10. B. R. Golla, et al., *Prog. Mater. Sci.* **111**, 100651 (2020).
11. A. S. Buyakov, Yu. A. Mirovoy, and S. P. Buyakova, *AIP Conf. Proc.* **2167**, 020046 (2019). doi 10.1063/1.5131913
12. R. Pei, et al., *J. Eur. Ceram. Soc.* **37**(2), 441–450 (2017).
13. A. Rezaie, et al., *Eur. Ceram. Soc.* **27**, 2495–2501 (2007).
14. A. G. Burlachenko, et al., *AIP Conference Proceedings* **2167**, 020042 (2019).
15. M. M. Opeka, et al., *J. Mater. Sci.* **39**(19), 5887–5904 (2004).
16. T. A. Parthasarathy, et al., *J. Am. Ceram. Soc.* **92**(5), 1079–1086 (2009).
17. E. J. Opila, et al., *Ceram. Eng. Sci. Proc.* **22**(3), 221–228 (2001).
18. W. B. Han, et al., *J. Am. Ceram. Soc.* **91**(10), 3328–3334 (2008).
19. R. Inoue, et al., *Ceram. Int.* **43**(11), 8081–8088 (2017).